# Bridge and size: the Kodaira–Spencer morphism, arithmetic derivatives and Step (xi) of IUT as lax 2-cells

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Version 1.0.0, 6 November 2026
doi:10.5281/zenodo.23984201

**Abstract**

In function fields, the height inequality (the analogue of Szpiro's inequality) follows from the Kodaira–Spencer morphism. This note places this classical argument, the "bridges" that are built in number fields from an additive relation $a + b = c$ (the $p$-adic logarithms of Baker's method and the arithmetic derivatives of Pasten), and Step (xi) of the proof of Corollary 3.12 of inter-universal Teichmüller theory (IUT) III side by side in the form of the lax 2-cells of the earlier note [SL]. Mochizuki himself describes the Kodaira–Spencer morphism as analogous to the "license to confuse" of IUT [Alien]. This note is an essay that proposes a viewpoint; it contains almost no new mathematical results.

There are two points. (1) The difficulty splits into a *bridge* (that a 2-cell exists) and a *size* (the blurring that must be allowed). This restates, in the language of [SL], two requirements stated by Mochizuki himself: a common container that is large enough to house both sides, and small enough to yield a nontrivial estimate [Roots]. In function fields both come automatically from the structure. A bridge from an additive relation exists unconditionally, but bounding its size is equivalent to the abc conjecture (Pasten [Pa]). In IUT, conversely, the blurring is small by the upper bound of IUT IV, while the bridge (the inclusion in Step (xi)) is not derived in the written argument. (2) A bridge from an additive relation does not cross the $\Theta$-link. The $\Theta$-link carries the unit portions by full poly-isomorphisms, so that specific global elements (such as $1 - \lambda$ of the Legendre form) are not carried. If one forces them across, a contradiction appears at the level of units already on actual initial $\Theta$-data (at the place of the datum D211 above 211, discrepancies of 87 and 232 levels for the labels 2 and 3). To absorb this discrepancy by blurring requires exactly half of the discrepancy at the level of value groups, which in the normalization of Corollary 3.12 (the $q$ side is measured by $\underline{\underline{q}} = q^{1/2\ell}$) is $\ell$ times the wall $(k-1)Q$. This grows with $q$ and does not fit under the upper bound of IUT IV. Moreover, the curves on which isogenies give a bridge (curves with a global multiplicative subspace) are excluded from the initial $\Theta$-data by the condition of Definition 3.1 (c) of IUT I. In summary, the reading of this note is that the ring structure (in particular, the addition) that IUT gives up in order to circumvent the absence of a global multiplicative subspace is precisely the material of a Kodaira–Spencer-type bridge (§8.1).

Hence, to complete IUT as an argument of Kodaira–Spencer type, one would have to build, inside the multiradial representation, a bridge whose blurring does not grow with $q$. For a single datum this claim is equivalent to the height inequality [SL], and if it holds for all data the abc conjecture follows [lana] (for arithmetic derivatives, the existence of small ones is equivalent to the abc conjecture [Pa]). The language of homotopy type theory (HoTT) can state this precisely, but cannot construct it. What HoTT contributed here is that, by requiring one to say along which path one identifies, it made visible that an additive bridge is information on the side of structure (§7.4). **No claim is made about whether IUT is correct.**

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# 1 Introduction

## 1.1 The question

P In earlier notes we saw that the dispute over Corollary 3.12 of IUT III concentrates on the point, in Step (xi) of its proof, that the value on the $q$ side is one of the possible values on the $\Theta$ side (below, "the inclusion"), and that this point is of the same strength as the height inequality. In the language of [SL], Corollary 3.12 is equivalent to a lax 2-cell between two paths, which is equivalent to $(k-1)Q \leq E$ [SL, Theorem 5.3]. Here $Q = |\log \underline{q}|$, $k = \ell(\ell+1)/12$ is the mean of $j^2$ over the labels $j = 1, \ldots, \ell^*$ (where $\ell^* = (\ell-1)/2$), and $E$ is the blurring allowed in the

comparison. In each of the readings listed in [RF], Corollary 3.12 and IUT IV together give the same height inequality.

H The question of this note is the following. In function fields, an inequality of the same shape follows structurally from the Kodaira–Spencer morphism. Can its "number-field version" be realized, as a quantity, on the construction of IUT?

### 1.2 What this note is

- **An essay that proposes a viewpoint.** Proposition 3.1 is classical, and Propositions 4.1, 4.2, 5.1, 7.1 and 7.2 are elementary. What this note adds is a dictionary that places three situations in one form, the dichotomy seen from it, and a concrete check that an additive bridge does not cross the Θ-link (end of §1.3).
- **What is not claimed.** We do not say that IUT is wrong, nor that the abc conjecture is false, nor that the inequality of Corollary 3.12 fails for any single datum (on the six completely factored data of this note it holds in the reading of the container [TC]). The undisputed parts of IUT (anabelian reconstruction, the étale theta function, Frobenioids) remain mathematics regardless of this note.
- **A "bridge" in this note is one brought in from outside.** What §7 shows is that a global relation lying outside the Θ-link, such as an additive relation, cannot be made to act on the comparison through the Θ-link. We do not exclude that a bridge of another form is built inside the multiradial representation (§8.2).

### 1.3 Prior work

- S **The analogy between the Kodaira–Spencer morphism and IUT is Mochizuki's own.** [Alien, §2.9] recalls the classical argument that derives the Szpiro inequality of function fields from the Kodaira–Spencer morphism, and [Alien, §3.1, (v)] reformulates it in terms of multiradiality and calls it analogous to the "license to confuse". [Alien, §2.3] describes the isogeny argument in the presence of a global multiplicative subspace and calls the "simulation" of such a subspace one fundamental theme of IUT. Mochizuki introduced an arithmetic Kodaira–Spencer morphism in 1999 [HA, Chapter IX], and in 2000 wrote that its Galois-theoretic version can only be defined when the Tate module has a natural rank one multiplicative subspace [GalKS].
- S **The two requirements on the container — large enough to house both sides, small enough to give a nontrivial estimate — are also stated by Mochizuki.** His slides on the classical roots of IUT give, as the key point of the Kodaira–Spencer argument, that $p_1^*\mathcal{E} \cong p_2^*\mathcal{E}$ serves as a common container, and say that one way to understand the multiradial representation of IUT III is as the construction of a common container that is "sufficiently large to house the incompatible ring structures on either side of the gluing constituted by the theta link $q_E^N \mapsto q_E$", but "sufficiently small to yield nontrivial estimate on the height of the elliptic curve over a number field under consideration" [Roots, §2]. In [EssLgc, §3.10, (ii), (iii)] he writes that "this approach consists of embedding $Y_1$ into larger and larger containers [i.e., the containers obtaining by adding the various indeterminacies] until the container becomes sufficiently large as to contain/engulf [not only $Y_1$, but also (!)] $Y_2$", and compares this with the PD-envelopes of the theory of crystals [EssLgc]. The "bridge" and the "size" of this note (Definition 2.1) restate these two requirements in the language of the model of [SL].
- X **That an arithmetic Kodaira–Spencer map is not yet available** is also stated in Fesenko's survey: "A (full) arithmetic version of the Kodaira–Spencer map could be quite useful for giving a proof of the arithmetic Szpiro conjecture. However, such an

arithmetic version of the Kodaira–Spencer map is not yet known." [Fe, §1.3]. In §1.4 he adds that "His earlier Hodge-Arakelov theory [23], where a certain weak arithmetic version of the Kodaira–Spencer map is studied, was already an innovative step forward" [Fe].

- [X] **A Kodaira–Spencer class for number fields does not exist, depending on the definition.** Faltings showed that a Kodaira–Spencer class for curves over number fields satisfying certain compatibility axioms cannot exist [Fal]; Dupuy–Zureick-Brown showed that, with slightly modified definitions, the Deligne–Illusie classes satisfy a compatibility condition of this kind [DZB].
- [X] **The equivalence of arithmetic derivatives adapted to an additive relation with the abc conjecture** is due to Pasten [Pa]. On $a+b=c$, arithmetic derivatives that are additive exist unconditionally, and the existence of sufficiently small ones is equivalent to the abc conjecture. Half of the view of this note, "the bridge is unconditional, the size is abc", is this result itself.
- [X] **A weak abc from linear forms in logarithms** is due to Stewart–Yu [StYu]; [Alien, §3.7, (iii)] compares it with IUT.
- [X] **The general theory of height inequalities and the Kodaira–Spencer map for function fields** is in M. Kim [Kim].
- [X] **Joshi** states that a paper treating Kodaira–Spencer theory in his theory of arithmetic Teichmüller spaces (Construction V) is in preparation [JoQA]. This note does not know its contents and does not use the outline in the questions and answers.
- [X] **That Step (xi) does not follow from the written properties** is shown in lana-agents/iut [lana] (`NonDerivable.lean`), by Higuchi [Hi] and in [SL, Theorem 5.7]. Scholze–Stix write of Mochizuki's claim that "up to the 'blurring' given by certain indeterminacies the diagram does commute; it seems to us that this statement means that the blurring must be by a factor of at least $O(\ell^2)$ rendering the inequality thus obtained useless" [SS, §2.2].
- [H] **What this note adds** is limited to three things. (a) A dictionary that places the Kodaira–Spencer morphism of function fields, the bridges from an additive relation (Baker, Pasten) and Step (xi) of IUT in the single form of a lax 2-cell of [SL]. (b) The assignment of the three situations by "bridge" and "size" (function fields have both, Baker and Pasten only the bridge, IUT only the size). The two requirements themselves are Mochizuki's (the item above); what this note adds is to state them as the existence of a lax 2-cell and its minimal blurring, and to put the additive bridges (Baker, Pasten) into the same table. That IUT has "only the size" is a reading of this note and points to the same place as Scholze–Stix, lana-agents/iut and Higuchi. We have not found this assignment in the sources we checked (Pasten, Dupuy–Zureick-Brown, Fesenko, Joshi's questions and answers, Mochizuki's papers and slides, lana-agents/iut, and a web search on 6 November 2026). (c) A check, along the definitions, that an additive bridge does not cross the $\Theta$-link, with numbers on actual data (§7). Item (c) is a concrete instance of the point of Scholze–Stix and of the incompatibility of the $\Theta$-link with the ring structure (arithmetic degrees) that Mochizuki himself describes ([R18, §14, (LbEx1)], [Alien, §3.3, (iv)]); it is not new in principle.

## 1.4 The shape of the argument

1. **Separate the bridge from the size** (§2). We recall the lax 2-cells of [SL] and define "bridge" and "size".

2. **In function fields both are automatic** (§3). The Kodaira–Spencer morphism is a lax 2-cell that exists automatically by the structure, with blurring $\deg \omega(\log)$.
3. **A bridge from an additive relation is unconditional; its size is abc** (§4). The $p$-adic logarithm bridge (Baker, Stewart–Yu) and the arithmetic derivatives of Pasten.
4. **The isogeny bridge lies outside the domain of IUT** (§5). Global multiplicative subspaces and Definition 3.1 (c).
5. **IUT has only the size** (§6). The statement of Theorem 3.11, where the source says the $\wedge$ is derived, and where elements specific to the curve enter.
6. **An additive bridge does not cross the Θ-link** (§7). What a full poly-isomorphism carries, the dilemma at the level of units, the language of HoTT, and what HoTT contributed.
7. **The dichotomy, and what a repair needs** (§8).

### 1.5 Labels

S a statement of the sources (IUT and Mochizuki's papers). X a claim of others (Pasten, Faltings, Dupuy–Zureick-Brown, Fesenko, lana-agents/iut, Scholze–Stix and others). P a statement proved in this note, or a fact shown in the earlier notes; a statement with only P is elementary. R the mathematics is known, and the way it is applied is this note's; the "known mathematics" is indicated. H a reading, interpretation or proposal of this note (not a proof).

## 2 Bridge and size: lax 2-cells

P We briefly recall the model of [SL, §5]. The value of the same object $b$ is carried to the real numbers along two paths. One is the path $\gamma$ through the Θ-link, which gives $\gamma_* b = -Q$ (the value on the $q$ side). The other is the path $c$ through the shared data, which gives $c_* b = -kQ$ (the value on the Θ side). The monodromy of the loop formed by the two paths is $1/k \neq 1$. ([SL] writes $\ell$ for the path $\gamma$ and $m = 1/k$ for the scale of the monodromy; we write $\gamma$ to avoid a clash with the prime $\ell$.)

- A strict 2-cell ($\gamma_* b = c_* b$) does not exist if $Q > 0$ [SL, Theorem 5.2].
- Corollary 3.12 (in the model) is equivalent to the lax 2-cell $\gamma_* b \leq c_* b + E$, which is equivalent to $(k-1)Q \leq E$ [SL, Theorem 5.3].
- If IUT IV bounds the blurring by $E \leq C$ ($C$ independent of $Q$), then Corollary 3.12 gives $Q \leq C/(k-1)$ [SL, Corollary 5.4].
- Qualitative properties such as IPL, SHE and APT (properties of 1-cells) do not decide whether a lax 2-cell exists [SL, Theorem 5.7].

**Definition 2.1** (Bridge and size)**.** H In a situation where values carried along two paths are compared, we call the assertion that a lax 2-cell exists for some blurring $E$ a *bridge*. When a bridge exists, we call the smallest blurring that makes it possible its *size*. A strict 2-cell is a bridge of size 0.

H Definition 2.1 restates, in the language of the model of [SL], the two requirements that Mochizuki states for the Kodaira–Spencer morphism and for the multiradial representation (that the common container be large enough to house both sides, and small enough to give a nontrivial estimate; §1.3). The bridge corresponds to the former and the size to the latter.

H To obtain the height inequality from Corollary 3.12, one needs both a bridge and an upper bound $E \leq C$ on its size. As we shall see, the three situations differ in which of the two they have.

# 3 Function fields: the Kodaira–Spencer morphism is a canonical lax 2-cell

S [Alien, §2.9] constructs the Kodaira–Spencer morphism, for a family of elliptic curves over a log curve $S^{\log}$ over $\mathbb{C}$, from the de Rham bundle $\mathcal{E}$, the Hodge subbundle $\omega_E \subseteq \mathcal{E}$ and the Gauss–Manin connection $\nabla_{\mathcal{E}}$, and reads it as follows.

> S "this diagram may be understood as a computation of the degree $\deg(-)$ of the 'positive mass' Frobenius-like object $\omega_E$ that yields an inequality" … "via a comparison of 'alien copies' of the 'positive mass' Frobenius-like object $\omega_E$ that lie on opposite sides of the 'link' constituted by a deformation of the moduli/holomorphic structure of the family of elliptic curves under consideration"
>
> Mochizuki, *The mathematics of mutually alien copies*, §2.9 [Alien]

Here the inequality is $2\cdot\deg(\omega_E) \le \deg(\omega_{S^{\log}/\mathbb{C}})$. [Alien, §3.1, (v)] restates the same argument in five steps $(1^{\mathrm{KS}})$–$(5^{\mathrm{KS}})$. One takes a line bundle $\mathcal{L}$ and a rank two bundle $\mathcal{F}$ with an exact sequence $0 \to \mathcal{L} \to \mathcal{F} \to \mathcal{L}^{-1} \to 0$. On the first infinitesimal neighbourhood $S^{\log}_{\delta}$ of the log diagonal, the isomorphism $p_1^*\mathcal{F} \simeq p_2^*\mathcal{F}$ of the two pull-backs given by the connection lets one compare the two pull-backs $p_1^*\mathcal{L}$ and $p_2^*\mathcal{L}$, with $\mathcal{F}$ as a "container" for $\mathcal{L}$. If the two images coincide, then $\deg\mathcal{L} = 0$ $((4^{\mathrm{KS}}))$; if not, the discrepancy gives a nonzero morphism of line bundles $\mathcal{L} \to \omega_{S^{\log}/\mathbb{C}} \otimes \mathcal{L}^{-1}$ $((5^{\mathrm{KS}}))$. For $\mathcal{F} = \mathcal{E}$ and $\mathcal{L} = \omega_E$ this is the Kodaira–Spencer morphism.

> S "the Kodaira-Spencer morphism may be thought of as a measure of the discrepancy that arises when one fixes the '$\omega_E$' on one factor of $S$ and compares it with the '$\omega_E$' on a distinct, 'alien' factor of $S$ by means of the common container '$\mathcal{E}$', which is equipped with a connection '$\nabla_{\mathcal{E}}$"' … "When formulated in this way, the Kodaira-Spencer morphism becomes manifestly analogous to the approach sketched in the Introduction to the present paper" … "i.e., [in the language of the Introduction] a 'license to confuse"'
>
> ibid., §3.1, (v) [Alien]

> S "in the limit '$N \to \infty$' [in which case comparison between '$h$' and '$N\cdot h$', or equivalently, between '$h$' and '$\frac{1}{N}\cdot h$', becomes a comparison between '$h$' and '0']"
>
> ibid., §3.1, (v) (on the relation between the classical argument and the argument of IUT that multiplies the height $h$ by $N$) [Alien]

**Proposition 3.1** (The Kodaira–Spencer morphism is a lax 2-cell)**.** R

*Known mathematics: classical ([Alien, §2.9, §3.1, (v)], [Kim]). The way it is applied: the 2-cells of [SL].*

*Let $S$ be a proper smooth curve over $\mathbb{C}$, $D \subseteq S$ a reduced divisor and $\omega = \Omega_S(\log D)$. Let $\mathcal{F}$ be a vector bundle of rank 2, $\nabla : \mathcal{F} \to \mathcal{F}\otimes\omega$ a logarithmic connection whose residues are nilpotent at every point, and $\mathcal{L} \subseteq \mathcal{F}$ a line subbundle with $\mathcal{F}/\mathcal{L} \simeq \mathcal{L}^{-1}$. For the composite*

$$\kappa : \mathcal{L} \hookrightarrow \mathcal{F} \xrightarrow{\nabla} \mathcal{F}\otimes\omega \to (\mathcal{F}/\mathcal{L})\otimes\omega = \mathcal{L}^{-1}\otimes\omega$$

*we have:*

*(i) $\kappa$ is $\mathcal{O}_S$-linear;*

*(ii) if $\kappa = 0$, then $\deg\mathcal{L} = 0$ (a strict 2-cell);*

*(iii) if $\kappa \neq 0$, then $2\deg\mathcal{L} \le \deg\omega = 2g - 2 + \#D$ (a lax 2-cell with blurring $\deg\omega$).*

*Proof.* (i) For $f \in \mathcal{O}_S$ and $s \in \mathcal{L}$, $\nabla(fs) = f\nabla s + s\otimes df$, and $s\otimes df$ lies in $\mathcal{L}\otimes\omega$, hence vanishes in $(\mathcal{F}/\mathcal{L})\otimes\omega$. (ii) If $\kappa = 0$, then $\nabla$ preserves $\mathcal{L}$ and gives a logarithmic connection on $\mathcal{L}$. Its residues are restrictions of nilpotent residues to an invariant line, hence 0, and the residue theorem gives $\deg\mathcal{L} = -\sum\mathrm{res} = 0$. (iii) There is a nonzero morphism of line bundles $\mathcal{L} \to \mathcal{L}^{-1}\otimes\omega$, so $\deg\mathcal{L} \le \deg\omega - \deg\mathcal{L}$. □

H In the language of [SL], the two pull-backs are the two paths, the connection is the shared data (the common container), the coincidence of the images is a strict 2-cell, and $\kappa$ is a lax 2-cell. **Both the bridge and the size come automatically from the structure.** The bridge comes from the fact that $\kappa$ is a morphism that is linear over the base ((i)). The size $\deg \omega$ is determined by the curve alone and does not depend on the height $\deg \mathcal{L}$ of the family. This linearity comes from the fact that the comparison takes place over a single base $S$ and its ring structure, and moreover in an infinitesimal neighbourhood. Number fields have neither derivations of this kind nor infinitesimal neighbourhoods (§4.5).

# 4 Number fields (1): bridges from an additive relation

## 4.1 The $p$-adic logarithm bridge

H In number fields, the closest thing to the discrepancy of Kodaira–Spencer is to linearize an additive relation by the $p$-adic logarithm. The exponent of a prime factor of one term (information in the value group) moves to the logarithm of the ratio of the other two terms (information in the unit group). In the words of Remark 3.12.2 (ii) of IUT III, this amounts to computing the value group portion in terms of the unit group portion; here this is done by an addition rather than by theta functions.

> S "this 'reassembling procedure' allows one to compute/estimate the value group portions of various monoids of arithmetic interest in terms of the unit group portions of these monoids. It is precisely these estimates that give rise to the inequality obtained in Corollary 3.12."
>
> IUT III, Remark 3.12.2, (ii) [IUT3]

**Proposition 4.1** (The additive bridge)**.** P

*Elementary. Numerical check:* `scripts/bridge_lib.py`.

*Let $a, b, c$ be coprime positive integers with $a + b = c$, $p$ an odd prime and $e = v_p(a) \geq 1$. Then:*

*(i)* $b/c \in 1 + p^e\mathbb{Z}_p$ *and* $v_p(b/c - 1) = e$;

*(ii)* $v_p(\log_p(b/c)) = e$;

*(iii)* $\log_p(b/c) = \sum_{q \mid bc} (v_q(b) - v_q(c)) \log_p q$, *that is,* $\log_p(b/c)$ *is a linear form in the $p$-adic logarithms of the primes dividing $bc$, with the exponents as coefficients.*

*The same holds for $a/c$ if $p \mid b$, and for $-a/b$ if $p \mid c$.*

*Proof.* (i) $b = c - a \equiv c \pmod{p^e}$ and $p \nmid c$ (coprimality), so $b/c - 1 = -a/c$ has valuation exactly $e$. (ii) If $p$ is odd and $v_p(x) = e \geq 1$, then for $n \geq 2$ we have $v_p(x^n/n) \geq ne - v_p(n) > e$ (as $v_p(n) \leq \log_3 n < n - 1$), so $\log(1 + x) = x - x^2/2 + \cdots$ has valuation $e$. (iii) $\log_p$ is a homomorphism on $\mathbb{Z}_p^\times$, and every prime dividing $bc$ is different from $p$. □

P We checked this on well-known abc triples of high quality, computing $\log_p$ from the series (Table 4.1, which lists all odd primes with exponent at least 2).

H For example, in $2 + 3^{10} \cdot 109 = 23^5$ the high power $3^{10}$ moves to the fact that the linear form $\log_3 2 - 5 \log_3 23$ in the 3-adic logarithms of 2 and 23 is unusually small 3-adically (valuation 10). The coefficients of the linear form are the exponents themselves (1 and 5), which are small.

## 4.2 Initial Θ-data in Legendre form

S In the application to ABC, IUT IV uses the line of parameters of elliptic curves in Legendre form.

Table 4.1: The additive bridge: the exponent $e$ in one term and the valuation of the $p$-adic logarithm of the ratio of the other two.

| triple $a+b=c$ | quality | $p^e$ | term | ratio | $v_p(\log_p(\text{ratio}))$ | primes in the ratio |
|---|---|---|---|---|---|---|
| $2+3^{10}\cdot 109=23^5$ | 1.6299 | $3^{10}$ | $b$ | $a/c$ | 10 | 2, 23 |
| | | $23^5$ | $c$ | $-a/b$ | 5 | 2, 3, 109 |
| $11^2+3^2\cdot 5^6\cdot 7^3=2^{21}\cdot 23$ | 1.6260 | $11^2$ | $a$ | $b/c$ | 2 | 2, 3, 5, 7, 23 |
| | | $3^2$ | $b$ | $a/c$ | 2 | 2, 11, 23 |
| | | $5^6$ | $b$ | $a/c$ | 6 | 2, 11, 23 |
| | | $7^3$ | $b$ | $a/c$ | 3 | 2, 11, 23 |
| $1+2\cdot 3^7=5^4\cdot 7$ | 1.5679 | $3^7$ | $b$ | $a/c$ | 7 | 5, 7 |
| | | $5^4$ | $c$ | $-a/b$ | 4 | 2, 3 |
| $283+5^{11}\cdot 13^2=2^8\cdot 3^8\cdot 17^3$ | 1.5808 | $5^{11}$ | $b$ | $a/c$ | 11 | 2, 3, 17, 283 |
| | | $13^2$ | $b$ | $a/c$ | 2 | 2, 3, 17, 283 |
| | | $3^8$ | $c$ | $-a/b$ | 8 | 5, 13, 283 |
| | | $17^3$ | $c$ | $-a/b$ | 3 | 5, 13, 283 |

S "We shall regard $X$ as the '$\lambda$-line"'

IUT IV, Corollary 2.2 [IUT4]

**Proposition 4.2** (The depth of the bridge on initial $\Theta$-data)**.** P

*Elementary.*

*Let $F$ be a number field, $\lambda \in F\setminus\{0,1\}$ and $E_\lambda : y^2 = x(x-1)(x-\lambda)$. Let $v$ be a place of odd residue characteristic at which $E_\lambda$ has (potentially) multiplicative reduction. In each of the cases $\mathrm{ord}_v(\lambda) = e > 0$, $\mathrm{ord}_v(1-\lambda) = e > 0$ and $\mathrm{ord}_v(\lambda) = -e < 0$, the element $1-\lambda$, $\lambda$, $\lambda/(\lambda-1)$ respectively is congruent to 1 modulo the $e$-th power of the prime ideal of $v$, and $\mathrm{ord}_v(q) = 2e$. Since these three cases are the only ones with multiplicative reduction at an odd place, at every odd bad place the depth of the complementary unit is $d_v = \mathrm{ord}_v(q)/2$.*

*Proof.* $j(E_\lambda) = 2^8(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$, and at a place of multiplicative reduction $\mathrm{ord}_v(q) = -\mathrm{ord}_v(j)$. In the three cases one counts $\mathrm{ord}_v(j) = -2e$ directly ($v$ is odd, so $2^8$ is a unit). The congruences follow from $1-\lambda = 1-$ (a small element) and so on. □

P The seven initial $\Theta$-data of this note are all in Legendre form over quadratic fields (the `data/` of [TC]). D211 has $F=\mathbb{Q}(\sqrt{5})$ and $\lambda=\pi^{29}$ ($\pi = 16+3\sqrt{5}$, of norm 211); at the place 211+ above 211, $\mathrm{ord}(q)=58$, that is, $1-\lambda \equiv 1 \pmod{\pi^{29}}$, of depth 29. The norm of $1-\lambda$ is $4\cdot 3^2\cdot 5\cdot 59\cdot 80621$ times two primes of 21 and 39 digits, and at the odd places dividing these the depth is 1 (the places above 2 are not bad places).

## 4.3 Baker's method and Stewart–Yu

X The smallness in Proposition 4.1 is the starting point of the method of linear forms in logarithms. Lower bounds for how $p$-adically small a linear form $\Lambda = \sum n_q \log_p q$ can be (the $p$-adic version of Baker's method) bound the exponent $e$ from above in terms of the primes dividing $bc$ and the size of the exponents. Stewart–Yu used estimates of this kind to prove a weak form of abc [StYu]. We have not checked their text; the shape of the conclusion is taken from the summary in [Alien].

> S "Recall that in [StYu], an inequality is obtained which may be thought of as a sort of 'weak version of the ABC Conjecture', i.e., which is, roughly speaking, weaker than the inequality of the usual ABC Conjecture in that it contains an undesired exponential operation 'exp(−)' in its upper bound."
>
> Mochizuki, *The mathematics of mutually alien copies*, §3.7, (iii) [Alien]

S Of this exponential discrepancy Mochizuki writes that "This sort of deviation from the inequality of the usual ABC Conjecture is of interest from the point of view of the 'vertical shift' discussed in §3.6, (iv)", and he identifies the vertical shift with a single iterate of the log-link in the log-theta-lattice (ibid.) [Alien].

H Here **the bridge is automatic**: $b/c \equiv 1 \pmod{p^e}$ follows from $a+b=c$ itself. Moreover, this bridge holds only on actual triples and distinguishes the curves (triples). What is weak is the **size**, which is determined by an estimate of transcendence theory, the lower bound for linear forms.

### 4.4 Pasten's arithmetic derivatives

X Pasten wrote this shape in the language of derivatives [Pa]. One takes the free module $\Omega$ generated by variables $\xi_p$, one for each prime, and the map $d:\mathbb{Z}\to\Omega$, $dn = n\sum_{p\mid n}(v_p(n)/p)\,\xi_p$, which satisfies the Leibniz rule. A group homomorphism $\psi:\Omega\to\mathbb{Z}$ with finite norm $\|\psi\| = \sup_p|\psi(\xi_p)|$ is called a derivation, and $d^\psi = \psi\circ d$. Let $\mathcal{T}(a,b)$ be the group of derivations with $\psi(\xi_p)=0$ for $p\nmid abc$ and $d^\psi a + d^\psi b = d^\psi c$. If the Wronskian $W^\psi(a,b) = a\,d^\psi b - b\,d^\psi a$ is nonzero, $a$ and $b$ are called $\psi$-independent.

- X $\mathcal{T}(a,b)$ has rank $r=\omega(abc)-1$, and by Siegel's lemma there are independent $\psi_1,\dots,\psi_r$ with $\prod_{i=1}^r\|\psi_i\| \le \frac{\omega(abc)}{2\log 2}\,c\log c$ (Theorem 2.6). For at least one of them $a,b$ are $\psi$-independent, and this $\psi$ satisfies $\|\psi\| \le \frac{\omega(abc)}{2\log 2}\,c\log c$ (Lemma 3.5).
- X If $a,b$ are $\psi$-independent, then $\frac{c}{\log c} \le \mathrm{rad}(abc)\cdot\frac{\|\psi\|}{\log 2}$ (Theorem 3.3; the same shape as Snyder's proof of Mason's theorem).
- X The small derivatives conjecture (Conjecture 3.9): for some $0<\eta<1$, for all triples except finitely many and those of the form $(1,N,q)$, there is $\psi\in\mathcal{T}(a,b)$ with $\|\psi\|<c^\eta$ that makes $a,b$ $\psi$-independent. It is equivalent to the abc conjecture, with a correspondence of exponents (Lemma 4.1, Theorem 4.5, Corollary 4.6; precisely, it follows from the abc conjecture of Masser–Oesterlé and implies the abc conjecture of Oesterlé with some exponent).

> X "the arithmetic analogue of the Mason-Stothers theorem is the abc Conjecture, but the proof for polynomials heavily uses derivatives and it is unclear how to adapt it to $\mathbb{Z}$." "In fact, we show that the existence of sufficiently small arithmetic derivatives in our sense is equivalent to the abc Conjecture."
>
> Pasten, §1.1 and abstract [Pa]

H In the language of 2-cells, $W^\psi(a,b)\neq 0$ is the bridge and $\|\psi\|$ is the size. **The bridge exists unconditionally (Lemma 3.5), and bounding its size is equivalent to the abc conjecture.** Putting the size $\frac{\omega}{2\log 2}c\log c$ of Lemma 3.5 into Theorem 3.3 gives only a trivial inequality. The derivative $d/dt$ of a function field lowers degrees by exactly 1, that is, it has "size 1", which is why the Mason–Stothers inequality is sharp.

### 4.5 Faltings's non-existence result and Dupuy–Zureick-Brown

> X "We show that an analog of the Kodaira–Spencer class for curves over number-fields cannot exist." "if $S$ is the spectrum of the ring of integers of a number field then there are no derivations and hence the Kodaira–Spencer map doesn't make sense." "By modifying his definitions slightly, we show that the Deligne–Illusie classes satisfy what could be considered an 'arithmetic Kodaira–Spencer' compatibility condition."
>
> Dupuy–Zureick-Brown, introduction (the first sentence is quoted there as the abstract of Faltings's paper) [DZB]

H The linearity of the function field argument (Proposition 3.1 (i)) came from derivations over the base. The ring of integers of a number field has none. Pasten avoids this by building additive "derivations" only on a single equation $a + b = c$, and in exchange loses control of the size. The Deligne–Illusie classes give another way around from the side of Frobenius lifts. In the language of this note, both are other devices for building a bridge.

## 5 Number fields (2): global multiplicative subspaces and isogenies

H There is a special case in which both a bridge and a size are available in number fields. S It is the case in which the $\ell$-torsion points of the elliptic curve have a Galois-stable subspace of rank one (a global multiplicative subspace).

> S "The quotient of the original given elliptic curve by such a global multiplicative subspace is an elliptic curve that is isogenous to the original elliptic curve. Moreover, the $q$-parameters of this isogenous elliptic curve are the $l$-th powers of the $q$-parameters of the original elliptic curve" ... "On the other hand, in general, such a global multiplicative subspace does not exist, and the issue of somehow 'simulating' the existence of a global multiplicative subspace is one fundamental theme of inter-universal Teichmüller theory."
>
> Mochizuki, *The mathematics of mutually alien copies*, §2.3 [Alien]

S According to the same place, the effect of the isogeny on differentials ($d\log(q) \mapsto l \cdot d\log(q)$) adds, at the level of heights, terms of the order of $\log(l)$, "thus giving rise to inequalities that are roughly of the form '$l \cdot h \leq h + \log(l)$"', and "this is precisely the approach that is taken to bounding heights in the 'tiny' special case of the theory of [Falt1] that is given in the proof of [GenEll], Lemma 3.5" [Alien] (there, [Falt1] is Faltings's paper of 1983 on finiteness theorems; see also [GenEll]). H Here the bridge is an algebraic map (compatible with the ring structure), the isogeny, and the size is of the order of $\log l$. [Alien, §3.1, (v)] also applies ($2^{\mathrm{KS}}$) and ($4^{\mathrm{KS}}$) of the Kodaira–Spencer argument to this situation.

> S "relative to the analogy with inter-universal Teichmüller theory, the situation that arises in ($2^{\mathrm{KS}}$), ($4^{\mathrm{KS}}$) corresponds to the [unusual!] situation in which there actually exists a 'global multiplicative subspace"'
>
> ibid., §3.1, (v) [Alien]

S Mochizuki's Galois-theoretic arithmetic Kodaira–Spencer morphism of 2000 can also be defined only where a multiplicative subspace exists.

> S "Unfortunately, however, this Lagrangian arithmetic Kodaira-Spencer morphism, which is based on a certain 'Lagrangian Galois action,' can only be defined when there is a natural (rank one) multiplicative subspace (i.e., 'weight 1' subspace) of the Tate module of the elliptic curve in question." ... "Ultimately, however, one would like to carry out this construction for elliptic curves over number fields."
>
> Mochizuki, *The Galois-theoretic Kodaira-Spencer morphism of an elliptic curve*, §0 [GalKS]

S IUT I regards the whole theory as a solution of the problem of constructing such a subspace, and states the price of "simulating" it.

> S "a sort of solution to the problem of constructing 'global multiplicative subspaces' and 'global canonical generators"' ... "the nonexistence of which in a 'naive, scheme-theoretic sense' constitutes the main obstruction to applying the theory of [HASurI], [HASurII] to diophantine geometry" ... "one can, in some sense, 'simulate' the notions of a 'global multiplicative subspace' or a 'global canonical generator'. On the other hand, such 'simulated global objects' may only be achieved at the cost of 'dismantling', or performing 'surgery' on, the global prime structure of the number fields involved"
>
> IUT I, introduction [IUT1]

S "this sort of operation of discarding certain of the primes of an NF can only be performed if one forgets the additive structure of an NF [i.e., since a sum of elements of an NF that are invertible at a given nonempty set of primes is no longer necessarily invertible at those primes!" ... "and works only with multiplicative structures, e.g., with Frobenioids."

Mochizuki, *The mathematics of mutually alien copies*, §3.3, (iv) [Alien]

S "$l$ is a prime number $\geq 5$ such that the image of the outer homomorphism $G_F \to \mathrm{GL}_2(\mathbb{F}_l)$ determined by the $l$-torsion points of $E_F$ contains the subgroup $\mathrm{SL}_2(\mathbb{F}_l) \subseteq \mathrm{GL}_2(\mathbb{F}_l)$"

IUT I, Definition 3.1, (c) [IUT1]

**Proposition 5.1** (The isogeny bridge is excluded from the initial Θ-data)**.** P

*Elementary.*

*If the condition of Definition 3.1 (c) holds, then $E_F[\ell]$ has no $G_F$-stable line. In particular, there is no global multiplicative subspace for this $\ell$.*

*Proof.* $\mathrm{SL}_2(\mathbb{F}_\ell)$ acts transitively on the nonzero vectors of $\mathbb{F}_\ell^2$, so it fixes no line. □

H The curves on which the isogeny bridge acts are excluded from the domain of IUT by definition. IUT deliberately treats the general case, in which no bridge of this kind is available. Moreover, in the application to ABC, $\ell$ is taken of the order of the square root of the height (IUT IV, Remark 2.2.4); even if a curve has an isogeny for some small $\ell$, that $\ell$ cannot be used.

S "The prime number $l$ is, ultimately, in the computations of Corollary 2.2, (ii) [cf., especially, condition '(C1)'], taken to be roughly of the order of the square root of the height of the elliptic curve under consideration."

IUT IV, Remark 2.2.4 [IUT4]

# 6 IUT: the blurring is small, but the bridge is not derived

## 6.1 The statement of Theorem 3.11

S Theorem 3.11 only fixes initial Θ-data as in Definition 3.1 of IUT I and adds no condition. Compatibility with the Θ-link is defined as stability, equivariance and functoriality with respect to automorphisms.

S "Fix a collection of initial Θ-data" ... "as in [IUTchI], Definition 3.1." ... "whose associated 'global degrees' may be computed by means of the log-volumes of (a)" ... "in the sense that these constructions are stabilized/equivariant/functorial with respect to arbitrary automomorphisms" [*sic*] "of the domain and codomain of these horizontal arrows of the LGP-Gaussian log-theta-lattice."

IUT III, Theorem 3.11, opening, (i)(c), (iii)(c) [IUT3]

P The statement of Theorem 3.11 does not contain the $q$-pilot. The $q$-pilot appears in Remark 3.11.1 (iii), immediately after, as the property (IPL) that the output is constructed in such a way that it is "linked/related" via full poly-isomorphisms to the $q$-pilot [IUT3].

S "This output data is constructed in such a way that it is linked/related, via full poly-isomorphisms of $\mathcal{F}^{\mathrm{I}\vdash\blacktriangleright\times\boldsymbol{\mu}}$-prime-strips induced by operations in the algorithm, to the input data prime-strip, i.e., the 'coric'/'fixed' $q$-pilot $\mathcal{F}^{\mathrm{I}\vdash\blacktriangleright\times\boldsymbol{\mu}}$-prime-strip, equipped with its rigidifying arithmetic holomorphic structure"

IUT III, Remark 3.11.1, (iii) (IPL) [IUT3]

P In Step (xi-f) the inclusion enters in the form that the construction of the subset of possible log-volumes of the output "is subject to the condition that this construction of output data

possibilities constitutes, in particular, a construction [perhaps only up to some sort of 'approximation', as a result of various indeterminacies] of the pilot-object log-volume of the input data" [IUT3], namely of $-|\log(\underline{q})|$; it does not follow from the written properties (IPL, SHE, APT) ([SL, Theorem 5.7], [lana], [Hi]). On the other hand, the upper bound $E \leq C$ on the blurring is given by the computation of IUT IV (the proof of Theorem 1.10), and the dispute lies on the side of Step (xi) of Corollary 3.12. **IUT has the size, but does not derive the bridge.**

### 6.2 Where the source says the $\wedge$ is derived

S The source itself names where the $\wedge$ of the comparison is derived from: the essentially tautological $\wedge$ coming from the fact that the two arithmetic holomorphic structures are distinguished ($(\mathrm{c}^{\mathrm{itw}})$) is mapped, through the subtle compatibility of the quotient operations of Remark 3.9.5 (viii), to the $\wedge$ on weak data that does not distinguish the two ($(\mathrm{f}^{\mathrm{itw}})$).

> S "data that is 'sufficiently weak' that one can no longer distinguish between $\Theta$-pilot and $q$-pilot objects." ... "it is essentially a tautological consequence of the fact that these two arithmetic holomorphic structures in the domain and codomain of the $\Theta^{\times\boldsymbol{\mu}}_{\mathrm{LGP}}$-link are distinguished from one another" ... "where the second '$\Longrightarrow$' of the above display is purely formal" ... "the '$\wedge$' of the above display may be regarded as the 'image' of, hence, in particular, as a consequence of, the '$\wedge$' in the display of $(\mathrm{c}^{\mathrm{itw}})$, via the various (sub)quotient operations discussed in Remark 3.9.5, (viii), i.e., whose subtle compatibility properties allow one to conclude the '$\wedge$' of the above display from the '$\wedge$' in the display of $(\mathrm{c}^{\mathrm{itw}})$"
>
> IUT III, Remark 3.12.2, (ii), $(\mathrm{b}^{\mathrm{itw}})$, $(\mathrm{c}^{\mathrm{itw}})$, $(\mathrm{f}^{\mathrm{itw}})$ [IUT3]

> S "the nontriviality of these various (sub)quotient operations lies in the very delicate balance between minimizing the indeterminacies that arise from passing to a quotient, while at the same time ensuring compatibility with the structures that exist prior to formation of the quotient." ... "forms a closed loop, i.e., up to the introduction of the 'formal quotient indeterminacies' discussed in (cQ3), (cQ4)" ... "a non-closed loop would yield a situation from which no nontrivial conclusions may be drawn" ... "passing to log-volumes [i.e., (sQ5)] amounts precisely to forgetting the local Galois groups and Frobenius-like units"
>
> IUT III, Remark 3.9.5, (viii), (ix) [IUT3]

H The closed loop closes at the level of structures that include the units and the local Galois groups of the prime-strips, before the passage (sQ5) to log-volumes. In the language of this note, the two sides of the "delicate balance" (keeping the indeterminacies small, and keeping compatibility with the original structures) correspond to the size ($E \leq C$) and the bridge (the inclusion). The source says that this balance is maintained, and the written text states it at the level of structures.

### 6.3 Where elements specific to the curve enter

H Information specific to the curve enters the construction as the "critical points" of $\kappa$-coric functions. S A critical point is a point lying over a closed point of the compactification $|C_L|^{\mathrm{cpt}}$ of the coarse space $|C_L|$ that arises from a 2-torsion point of $E_F$; those not over the cusp are called strictly critical (for the Legendre form, those over $x = 0, 1, \lambda$ and the cusp $\infty$).

> S "We shall refer to a closed point of the proper smooth curve determined by some finite subextension" ... "as a critical point if it maps to a closed point of the [proper smooth] compactification" ... "that arises from one of the 2-torsion points of $E_F$" ... "$f$ restricts to a root of unity at every strictly critical point of $|C_L|^{\mathrm{cpt}}$" ... "restriction to the strictly critical points determines a sort of canonical splitting up to roots of unity" ... "of the set of nonzero constant [i.e., $L$-] multiples of $\kappa$-coric functions into a direct product, up to roots

> of unity, of the set of $\kappa$-coric functions and the set of nonzero elements of $L$" ... "since $|C_L|^{\mathrm{cpt}}$ has precisely 4 critical points"
>
> IUT I, Remark 3.1.7, (i), (ii) [IUT1]

S Restricting the Kummer classes of these functions to the decomposition groups of non-critical points, one recovers the number field $F_{\mathrm{mod}}$ as a field (IUT I, Example 5.1 (v): "together with the field structure — and hence, in particular, the topologies determined by the valuations —" on the union of the monoids with $\{0\}$) [IUT1]. IUT II, Corollary 4.8 states the compatibility of the isomorphisms of this Kummer theory with the labels, the $\mathbb{F}_l^*$-symmetry and localization, and its proof is "The various assertions of Corollary 4.8 follow immediately from the definitions and the references quoted in the statements of these assertions." [IUT2]. In Theorem 3.11 (i)(c), the copies of the number field are placed in the product of log-shells, and their role is to compute global degrees by log-volumes (the quotation in §6.1).

**Observation 6.1** (Elements specific to the curve enter the construction, not the comparison)**.** H

*A reading on the statements of Theorem 3.11, IUT II Corollary 4.8, IUT I Example 5.1 and Remark 3.1.7.*

$\lambda$ enters the construction as a tool for recovering the number field $F_{\mathrm{mod}}$ rigidly (the critical points of $\kappa$-coric functions, the canonical splitting of the constants). The only global information that enters the comparison (the inequality between log-volumes) is the global degree of arithmetic line bundles of the number field, that is, the sum of local log-volumes (the product formula). The product formula holds for every element and so does not distinguish curves. No statement in these theorems makes the additive relation $\lambda + (1-\lambda) = 1$ act on the comparison as a quantity.

H Applying the product formula to $j(E)$ gives only an identity (the contribution of $q$ at the bad places equals the contribution of the zeros of $j$ plus the archimedean contribution), not an upper bound. To make it an upper bound one needs to know how the additive connection acts as a quantity, and that is the abc conjecture itself (§4).

# 7 An additive bridge does not cross the Θ-link

## 7.1 What the Θ-link carries

S The Θ-link carries the unit portions by full poly-isomorphisms, and the categories glued together are multiplicative.

> S "the full poly-isomorphism of $\mathcal{F}^{\vdash\times\boldsymbol{\mu}}$-prime-strips" ... "it is of crucial importance that the categories that are glued together via the $\Theta^{\times\boldsymbol{\mu}}_{\mathrm{LGP}}$-link be purely multiplicative in nature"
>
> IUT III, Theorem 3.11, (iii)(a), and Remark 3.9.5, (x) [IUT3]

**Proposition 7.1** (What a full poly-isomorphism carries)**.** P

*Elementary.*

*Let $X, Y$ be objects of a category, $\Phi = \mathrm{Isom}(X, Y)$ the full poly-isomorphism and $T$ a functor that assigns a set to each object. For $x \in T(X)$, the set $\{T(\varphi)(x) \mid \varphi \in \Phi\}$ obtained by carrying $x$ along $\Phi$ is one $\mathrm{Aut}(Y)$-orbit. It is a single point if and only if $x$ is fixed by $\mathrm{Aut}(X)$. For a property $P$ invariant under isomorphisms, $P(X) \iff P(Y)$.*

*Proof.* For $\varphi, \varphi' \in \Phi$, $\varphi' = (\varphi'\varphi^{-1})\varphi$, and $\varphi'\varphi^{-1}$ runs over all of $\mathrm{Aut}(Y)$. The last assertion is the definition. □

H The material of an additive bridge consists of the units $u_v = (1-\lambda)_v$ (or their logarithms) and of the consistency that "they all come, at every bad place, from one and the same global element". The former are specific elements of the unit portion, and, for instance, the

automorphisms (Ism) of (Ind2) move $\log u_v$ to other elements. By Proposition 7.1, therefore, what the $\Theta$-link can carry is a property of the type "there is a unit of depth $d_v$", not the element $1-\lambda$. The latter consistency is not contained in the data of the local $\mathcal{F}^{\vdash\times\mu}$-prime-strips, and the global portion only has the realified degrees. The source itself writes that these data are "'sufficiently weak' that one can no longer distinguish between $\Theta$-pilot and $q$-pilot objects" (§6.2) [IUT3].

## 7.2 The dilemma at the level of units

**Proposition 7.2** (Keeping an additive bridge on both sides of the $\Theta$-link). P

*Elementary. Numbers:* `scripts/bridge_lib.py`.

*Let $\mathbb{V}^{\mathrm{bad}}$ be the same set of bad places as in [TC] (the places with $\mathrm{ord}_v(q) > 0$ that are not above 2, 3, 5 or $\ell$ and at which $\mathrm{ord}_v(q)$ is not divisible by $\ell$), and let $d_v = \mathrm{ord}_v(q)/2$ be the depth of the additive bridge at $v \in \mathbb{V}^{\mathrm{bad}}$ (Proposition 4.2). The $\Theta$-link identifies the $\Theta$-pilot $\underline{\underline{q}}^{j^2}$ of label $j$ with the $q$-pilot $\underline{\underline{q}}$ on the other side, so that, on taking $2\ell$-th powers, $q^{j^2}$ on this side corresponds to the $q$ of the curve on the other side. Suppose that (a) the units of a bridge of depth $d_v$ are carried across the $\Theta$-link, (b) the carried units are required to be an additive bridge on the other side as well, that is, units of depth $\frac{1}{2}\mathrm{ord}_v(q^{j^2}) = j^2 d_v$ for the $q$ of the other side (which corresponds to $q^{j^2}$ on this side), and (c) the operation of carrying changes the depth by at most $\varepsilon_{v,j}$ levels. Then $(j^2-1)\,d_v \le \varepsilon_{v,j}$, and averaging over the labels and summing with the weights $f_v \log p_v/[F:\mathbb{Q}]$ gives*

$$\frac{k-1}{2}\cdot\frac{1}{[F:\mathbb{Q}]}\sum_{v\in\mathbb{V}^{\mathrm{bad}}}\mathrm{ord}_v(q)\,f_v\log p_v \;\le\; \frac{1}{[F:\mathbb{Q}]}\sum_{v\in\mathbb{V}^{\mathrm{bad}}}\bar{\varepsilon}_v\,f_v\log p_v,$$

*where $\bar{\varepsilon}_v$ is the average of $\varepsilon_{v,j}$ over the labels. The left-hand side is exactly half of the discrepancy at the level of value groups (the average difference between $q$ and $q^{j^2}$) measured with the same weights. In the normalization of Corollary 3.12,*

$$Q = |\log \underline{\underline{q}}| = \frac{1}{2\ell[F:\mathbb{Q}]}\sum_{v\in\mathbb{V}^{\mathrm{bad}}}\mathrm{ord}_v(q) f_v\log p_v,$$

*the left-hand side is $\ell\,(k-1)Q$, that is, $\ell$ times the wall $(k-1)Q$.*

*Proof.* By (c) the depth of the carried units differs from $d_v$ by at most $\varepsilon_{v,j}$, and by (b) it must be $j^2 d_v$, so $(j^2-1)d_v \le \varepsilon_{v,j}$. The average of $j^2$ over the labels $j = 1,\dots,\ell^*$ is $k$, so the average of $j^2-1$ is $k-1$. Insert $d_v = \frac{1}{2}\mathrm{ord}_v(q)$ and sum. The last equality follows from the formula for $Q$. □

P A unit multiple coming from (Ind2) does not change the depth ($\varepsilon_{v,j} = 0$). Hence keeping (a) and (b) leads to a contradiction on every datum with a bad place. This happens on actual initial $\Theta$-data (Table 7.1).

H Hence there are only two ways to repair IUT by bringing in an additive bridge from outside. If the bridge is kept on both sides, the contradiction of Scholze–Stix appears at the level of units. If it is carried without being kept, the information of the bridge (which element, and its global consistency) is lost. If one takes the middle way and absorbs the discrepancy by blurring, the blurring needed is $\ell$ times the wall and grows with $Q$. The upper bound $C$ of IUT IV does not contain this blurring, so the comparison becomes trivial and says nothing about the height. This is the same shape as Scholze–Stix's blurring "rendering the inequality thus obtained useless" [SS, §2.2].

Table 7.1: The depth of the additive bridge, and the discrepancy when one tries to keep it on both sides.

| datum | $\ell$ | place of the largest bridge | ord($q$) | depth $d$ | gap at $j=2$ | gap at $j=\ell^*$ | sum of gaps (LHS) | wall $(k-1)Q$ |
|---|---|---|---|---|---|---|---|---|
| D211 | 7 | 211+ | 58 | 29 | 87 | 232 ($j=3$) | 559.56 | 79.94 |
| D109 | 7 | 109+ | 58 | 29 | 87 | 232 ($j=3$) | 490.60 | 70.09 |
| D2 | 7 | a 50-digit prime (−) | 2 | 1 | 3 | 8 ($j=3$) | 440.17 | 62.88 |
| D3 | 7 | 30965771− | 60 | 30 | 90 | 240 ($j=3$) | 2,068.45 | 295.49 |
| D18 | 7 | 30965771+ | 36 | 18 | 54 | 144 ($j=3$) | 1,127.41 | 161.06 |
| D15 | 7 | 30965819+ | 30 | 15 | 45 | 120 ($j=3$) | 946.12 | 135.16 |
| D2357† | 157 | 2357+ | 944 | 472 | 1,416 | 2,871,176 ($j=78$) | 3,840,837.03 | 24,463.93 |

"Place of the largest bridge": the place of $\mathbb{V}^{\mathrm{bad}}$ at which $d_v f_v \log p_v$ is largest. The gaps are numbers of levels (differences of depth). The left-hand side and the wall are in the normalization of Proposition 7.2; their ratio is exactly $\ell$ on every datum. †D2357 does not count the places above an unfactored cofactor, so it is conditional (on this datum the wall exceeds $C$ [TC]).

## 7.3 In the language of HoTT

H In the language of [SL]: what can be carried along a path of identification is a property (a proposition), and the simplification of Scholze–Stix is always correct for propositions, by path induction. If a structure (a chosen element) is carried along an identification consisting of "all isomorphisms", only an orbit of the automorphism group is determined (Proposition 7.1). An additive bridge is information on the side of structure, so it cannot become a lax 2-cell that crosses the Θ-link. In function fields the Kodaira–Spencer morphism could be a bridge because the comparison was made by a connection over a single base $S$, that is, by a specific isomorphism $p_1^*\mathcal{F} \simeq p_2^*\mathcal{F}$, compatible with the ring structure (Proposition 3.1 (i)). Mochizuki also describes the incompatibility of the Θ-link with arithmetic degrees as the monodromy of a loop.

> S "Alternatively, one may understand the incompatibility of the Θ-link (i.e., a horizontal arrow in the log-theta-lattice) with arithmetic degrees (cf. (AD)) as a sort of nontrivial monodromy around the loop (i.e., a special case of '$\Gamma_{\mathrm{loop}}$'!) that arises as soon as one identifies the domain and codomain of the Θ-link — an incompatibility, i.e., 'contradiction', that is in fact avoided in IUTch (cf. (IUAD)) precisely by distinguishing the (ring structures in the) domain and codomain of the Θ-link, i.e., by working with '$\Gamma_{\mathrm{seg}}$', and then applying the multiradial algorithms of IUTch to obtain an alternative way to compute this nontrivial monodromy (cf. (GIUT))."
>
> Mochizuki, *Report on discussions … March 15–20, 2018*, §14, (LbEx1) [R18]

## 7.4 What HoTT contributed

H Every proposition of this note can be proved without HoTT, and we wrote them in ordinary mathematics. Proposition 7.1 is an elementary lemma of category theory, and Mochizuki's full poly-isomorphisms and indeterminacies are this idea from the start. The language of HoTT was nevertheless useful, because it puts the identification, down to "along which path", into the types. In HoTT an identification is itself a datum, a path, and a proposition cannot be written down without naming the path. From an identification that does not choose a path (all isomorphisms), only properties can be extracted. Applying this distinction, one sees at once that an additive bridge is information on the side of structure and does not cross the Θ-link. Since the dispute over IUT is about how mutually alien copies are identified, a type discipline of exactly this kind applies. [SL, §1.1] gave three reasons for choosing HoTT: Mochizuki himself explains the dispute by segment, loop, universal cover and monodromy; an identification is a path, a first-class object; and univalence decides, as a theorem, when identifying isomorphic objects is harmless. This note is one example in which those reasons applied.

H There are limits as well. We used only the low levels of HoTT (propositions, sets, groupoids), at which the statements of HoTT translate directly into the language of orbits, torsors and groupoids. The advantage of HoTT is that it makes the discipline of groupoids the default and enforces it, not that it can say what other languages cannot. The claims at the level of HoTT are stated on paper and have not been checked by machine (the verification in Lean in [SL] covers only the quantitative core, and the equality of Lean is incompatible with univalence). This is one case, and a case favourable to HoTT, since identification itself is what is disputed. One can say that HoTT has produced new mathematics when a theorem comes out that would have been hard to find without it; for instance, Anel, Biedermann, Finster and Joyal generalized the Blakers–Massey theorem to an arbitrary higher topos and an arbitrary modality and write that "Our proof is inspired by the one given in Homotopy Type Theory" [ABFJ]. This note contains no result of that kind.

# 8 The dichotomy, and what a repair needs

## 8.1 Table

Table 8.1: Bridge and size (the reading of this note).

| situation | bridge (that a 2-cell exists) | size (blurring) | shape of the conclusion |
|---|---|---|---|
| function fields (Kodaira–Spencer) | automatic from the structure ($\kappa$ is a linear morphism) | $\deg \omega(\log)$, determined by the curve alone | $2 \deg \omega_E \leq \deg \omega(\log)$ |
| curves over number fields with a global multiplicative subspace | the isogeny (an algebraic map) | of the order of $\log l$ | of the form $l \cdot h \leq h + \log l$ ([Alien, §2.3]); excluded in IUT by Definition 3.1 (c) (Proposition 5.1) |
| Baker's method, Stewart–Yu | automatic from the addition (Proposition 4.1) | determined by the lower bound for linear forms in logarithms, large | a weak abc (a shift in the exponent) |
| Pasten's arithmetic derivatives | exists unconditionally (Lemma 3.5) | $\|\psi\|$; small $\iff$ abc (Corollary 4.6) | $c/\log c \leq \mathrm{rad}(abc)\|\psi\|/\log 2$ |
| IUT (Step (xi) of Corollary 3.12) | not derived in the written argument (§6) | small, $E \leq C$ (IUT IV) | $(k-1)Q \leq E$ |

H In one sentence, the reading of this note is the following. **IUT gave up the ring structure (in particular, the addition) in order to circumvent the absence of a global multiplicative subspace, but that ring structure was precisely the material of a Kodaira–Spencer-type bridge.** The bridge of function fields came from a specific isomorphism over a single base, compatible with the ring structure (Proposition 3.1 (i), §7.3). The introduction of IUT I names, as the price of simulating a global multiplicative subspace, the "dismantling" of the global prime structure of the number fields involved [IUT1], and according to [Alien, §3.3, (iv)] the operation of discarding primes can only be performed if one forgets the additive structure (the quotations in §5). The curves on which the isogeny bridge acts are excluded by Definition 3.1 (c) (Proposition 5.1), and an additive bridge does not cross the $\Theta$-link, which glues multiplicative data by full poly-isomorphisms (Propositions 7.1 and 7.2). What remains is the inner bridge of §8.2. This is a reading, not a theorem.

## 8.2 What a repair needs

H To complete IUT as an argument of Kodaira–Spencer type, one needs the following.

- **A bridge built inside the multiradial representation.** The way of bringing in an additive relation from outside is closed (Propositions 7.1 and 7.2). The isogeny bridge is excluded by Definition 3.1 (c) (Proposition 5.1).
- **That its blurring does not grow with** $q$ ($E \leq C$), uniformly for all initial $\Theta$-data.

H For a single datum this claim is equivalent to the height inequality ([SL, Theorem 5.3]), and if it holds for all genuine initial $\Theta$-data, the abc conjecture follows (`classicalABC_of_forall_xi` of [lana]; for arithmetic derivatives, the equivalence of the existence of small ones with the abc conjecture is [Pa, Corollary 4.6]). The language of HoTT can state the required 2-cell precisely, but cannot construct it ([SL, Theorem 5.7]). That is, the repair amounts not to an adjustment of IUT but to finding a new proof of the height inequality. One form of it would be to show the (FrgInv) of the source, "the operation of forgetting discussed in (MnAlyLb) can in fact, if one allows for suitable indeterminacies, be inverted" [EssLgc, §3.11], at the level of values rather than of structures, with an indeterminacy independent of $q$.

S Anabelian reconstruction itself constructs a number field group-theoretically from a fundamental group. Theorem 1.9 of [AbsTopIII] reconstructs, for a curve over a sub-$p$-adic field $k$, the NF-portion of the function field from the extension $1 \to \Delta_X \to \Pi_X \to G_k \to 1$. Remark 1.9.4 says that, for a local field $k$ (an MLF), of the two dimensions of $G_k$ the one represented by the inertia group is not rigid, but that both dimensions are rigidified once coupled with $\Delta_X$.

> S "there exists a functorial 'group-theoretic' algorithm" … "for reconstructing the 'NF-portion of the function field' of $X$ from the extension of profinite groups $1 \to \Delta_X \to \Pi_X \to G_k \to 1$" … "On the other hand, the dimension of $G_k$ represented by the inertia group in $I_k \subseteq G_k$ [which, as is well-known, is of cohomological dimension 1] is 'far from rigid"' … "The 'group-theoretic' algorithm of Theorem 1.9 shows that the condition of being 'coupled with $\Delta_X$' [i.e., via the extension determined by $\Pi_X$] has the effect of rigidifying both of the 2 dimensions of $G_k$"
>
> Mochizuki, *Topics in absolute anabelian geometry III*, Theorem 1.9 and Remark 1.9.4 [AbsTopIII]

H This reconstruction works inside one theater (where $\Pi_X$ is available) and works in the same way for every curve. lana-agents/iut contains a formalization in Lean of the steps of this reconstruction that recover the additive structure (Uchida's lemma, Proposition 1.3, and Theorem 1.9 (e); `Iut/Reconstruction/`) [lana]. Across the $\Theta$-link, only the Galois group $G_v$ and the unit portions (not $\Pi_v$) are carried locally, and only the realified degrees globally; so the "coupling" of Remark 1.9.4 is not available, and the addition returns only up to the indeterminacy of (Ind2), as we read it. What a repair needs is a **quantitative version** of this reconstruction (a claim, measured by log-volumes, of how precisely the addition returns after crossing the $\Theta$-link). It is not written.

### 8.3 Rescue in a weak sense

- H **A weak Corollary 3.12 with a large blurring.** If one allows the blurring of the size of Pasten or Stewart–Yu, a weak form of the comparison holds. But that is only a restatement of existing theorems.
- H **Curves with a global multiplicative subspace.** Isogenies give a bridge. But, as Proposition 5.1 shows, these curves lie outside the domain of IUT.
- H **Function fields.** The Kodaira–Spencer morphism gives a bridge (Proposition 3.1). This is classical, and Mochizuki himself writes it as the limiting case $N \to \infty$ of the argument (of the type of IUT) that multiplies the height by $N$ ([Alien, §3.1, (v)]).

H None of these adds new arithmetic. As another kind of rescue, there is the view that Corollary 3.12 is not "nonsense" but an exact proposition, a lax 2-cell, whose truth is merely unresolved ([SL]).

# 9 Limitations

- **An essay that proposes a viewpoint.** Proposition 3.1 is classical, and Propositions 4.1, 4.2, 5.1, 7.1 and 7.2 are elementary; there is no new theorem. The dichotomy of Table 8.1 is an analogy, not a theorem. The correspondence between function fields and number fields follows Mochizuki's analogy [Alien] and the analogy of Vojta and Pasten [Pa].
- **Section 7 is about bridges brought in from outside.** It is a check along the definition of the Θ-link (full poly-isomorphisms, multiplicative gluing); it does not show that there is no construction that carries global consistency in another form inside the multiradial representation. If there were one, it would be the repair of §8.2.
- **Condition (b) of Proposition 7.2 is not something IUT requires.** IUT places no additive relation on the Θ side. What Proposition 7.2 shows is only that a repair bringing in an additive relation from outside is either contradictory or needs a blurring that grows with $Q$.
- **Observation 6.1 is a reading on the statements of theorems.** We read the statements of Theorem 3.11, IUT II Corollary 4.8, IUT I Example 5.1 and Remark 3.1.7, but not all of the proofs and the other remarks.
- **Literature.** We have not read the papers of Faltings, Stewart–Yu and Kim, [GenEll], or the text of Baker's method themselves (we rely on the introduction of Dupuy–Zureick-Brown and on the bibliographies and summaries of Pasten and [Alien]). Joshi's Construction V is stated to be in preparation, and we have not confirmed its existence or contents. The work of Anel–Biedermann–Finster–Joyal is cited from its abstract.
- **Data:** four abc triples and seven initial Θ-data; D2357 is conditional. Proposition 4.2 is a statement about places of multiplicative reduction, and at all places counted $\mathrm{ord}(q)$ was even (checked by the script).
- **The part on HoTT** (§7.3, §7.4) depends on the framework and reading of [SL]; the assessment of §7.4 is about this one case.
- No claim is made about whether IUT is correct.

# 10 Checks and reproduction

- `scripts/bridge_lib.py` computes Table 4.1 (four triples: the valuation of the $p$-adic logarithm, computed from the series, agrees with the exponent) and Table 7.1 (seven data: $\mathrm{ord}(q)$ is even, the gaps, and the label-averaged sum is $\ell$ times the wall). The values of the wall and of $Q$ agree with the output of the programs of [TC]. Its output is `outputs/bridge.txt` (`python3 scripts/bridge_lib.py > outputs/bridge.txt`); `paper/make_generated.py` writes the tables and the numbers of the text from the same functions.
- Data: `data/D*.txt`, the same files as in `data/` of [TC] (version 1.0.0).
- Quotations are taken from the English originals (the text layers of the PDFs, with mathematical symbols and decorations checked on the page images).

## Use of generative AI

The question of this note — whether the number-field version of the Kodaira–Spencer argument, which gives the height inequality for function fields, can be realized as a quantity on the construction of IUT — and the decision to write it up as a separate note arose in the author's discussions with generative AI tools; the author judged them important and decided to make them the subject of this note. Generative AI tools (large language models) were used, under the author's direction, in the computations, in checking the literature and in the writing. The author takes full responsibility for the contents of this note.

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